

# Project Design Phase

Project Design Template

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## Team Members

Mangalagiri Tejaswi — Team Lead

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## Cab Booking System

Maximum Marks: 5 Marks

## Problem Statement

Many users face difficulties in booking cabs quickly, checking cab availability, and managing ride details through traditional methods. There is also a need for administrators to efficiently manage users, bookings, and cab information in a centralized system.

## Proposed Solution

The Cab Booking System is a MERN Stack web application that enables users to register, log in, search for available cabs, book rides, manage bookings, and view ride history. Administrators can manage users, bookings, and cab details through a dedicated dashboard. The system provides secure authentication, responsive design, and efficient booking management.

## Objectives

Develop a secure cab booking platform.

Enable users to register and log in securely.

Allow users to book and manage cab rides.

Display available cabs and booking history.

Provide an admin dashboard for system management.

Ensure responsive design across devices.

## Functional Requirements

User Registration

User Login

Cab Booking

View Available Cabs

Booking History

Ride Management

Admin Dashboard

Secure Authentication

## Non-Functional Requirements

Fast response time

Secure user authentication

Responsive UI

Reliable database storage

Easy navigation

High availability

## Technology Stack

Frontend

HTML

CSS

JavaScript

React.js

Bootstrap

Backend

Node.js

Express.js

Database

MongoDB

Tools

Visual Studio Code

Git

GitHub

Postman

## High Level Architecture

User

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React Frontend

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Express.js API

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Node.js Server

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MongoDB Database

## Modules

User Authentication

Cab Booking

Ride Management

Booking History

Admin Dashboard

Database Management

## Expected Outcome

The project provides a secure and user-friendly platform for booking cabs online. Users can easily manage rides, while administrators efficiently monitor bookings, users, and cab information through a centralized dashboard.